

Homework1

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1 AI Use and Verification Statement

I used ChatGPT (OpenAI, GPT-6; accessed September 29, 2026) to help draft this AI-use statement. I checked the statement against the homework instructions. ChatGPT has not been used to generate the answers or R calculations reported in this submission.

2 Question 1

2.1 Question 1.1

Let $G_1 = \{j : X_j = 0\}$ and $G_2 = \{j : X_j = 1\}$, where $\sum_{j \in G_1} = n_1$ and $\sum_{j \in G_2} = n_2$. The model gives

$$\mu_1 = \beta_0, \quad \mu_2 = \beta_0 + \beta^*.$$

$$H_0 : \mu_1 = \mu_2 \equiv H_0 : \beta^* = 0.$$

$$L(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2) = \prod_{j \in G_1} \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left\{ -\frac{(Y_j - \beta_0)^2}{2\sigma_1^2} \right\} \prod_{j \in G_2} \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp \left\{ -\frac{(Y_j - \beta_0 - \beta^*)^2}{2\sigma_2^2} \right\}.$$

$$\ell(\beta_0, \beta^*, \sigma_1^2, \sigma_2^2) = -\frac{n_1 + n_2}{2} \log(2\pi) - \frac{n_1}{2} \log(\sigma_1^2) - \frac{n_2}{2} \log(\sigma_2^2) - \frac{1}{2\sigma_1^2} \sum_{j \in G_1} (Y_j - \beta_0)^2 - \frac{1}{2\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*)^2.$$

Treating σ_1^2 and σ_2^2 as known, the score equations are

$$\begin{aligned} \frac{\partial \ell}{\partial \beta_0} &= \frac{1}{\sigma_1^2} \sum_{j \in G_1} (Y_j - \beta_0) + \frac{1}{\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*), \\ \frac{\partial \ell}{\partial \beta^*} &= \frac{1}{\sigma_2^2} \sum_{j \in G_2} (Y_j - \beta_0 - \beta^*). \end{aligned}$$

Setting the scores to zero gives

$$\hat{\beta}_0 = \bar{Y}_1, \quad \hat{\beta}^* = \bar{Y}_2 - \bar{Y}_1.$$

The negative Hessian, which is also the Fisher information for $(\beta_0, \beta^*)^T$, is

$$I = \begin{pmatrix} \frac{n_1}{\sigma_1^2} + \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \\ \frac{n_2}{\sigma_2^2} & \frac{n_2}{\sigma_2^2} \end{pmatrix}.$$

With the equation of inversion

$$V = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}, \quad ac - b^2 \neq 0.$$

$$I^{-1} = \begin{pmatrix} \frac{\sigma_1^2}{n_1} & -\frac{\sigma_1^2}{n_1} \\ -\frac{\sigma_1^2}{n_1} & \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \end{pmatrix}.$$

Thus Wald statistic is

$$Z_W = \frac{\hat{\beta}^*}{\sqrt{\text{Var}(\hat{\beta}^*)}} = \frac{\bar{Y}_2 - \bar{Y}_1}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

$$W = Z_W^2 = \frac{(\bar{Y}_2 - \bar{Y}_1)^2}{\sigma_1^2/n_1 + \sigma_2^2/n_2}.$$

2.2 Question 1.2

$$S_g = \sum_{j \in G_g} (Y_j - \bar{Y}_g)^2, g = 1, 2.$$

$$\hat{\beta}_0 = \bar{Y}_1, \quad \hat{\beta}^* = \bar{Y}_2 - \bar{Y}_1, \quad \hat{\sigma}_{g,\text{ML}}^2 = \frac{S_g}{n_g}.$$

$$\begin{aligned} W_{\text{ML}} &= \frac{(\hat{\beta}^*)^2}{\hat{\sigma}_{1,\text{ML}}^2/n_1 + \hat{\sigma}_{2,\text{ML}}^2/n_2} \\ &= \frac{(\bar{Y}_2 - \bar{Y}_1)^2}{S_1/n_1^2 + S_2/n_2^2}. \end{aligned}$$

plug in:

$$s_g^2 = \frac{S_g}{n_g - 1}.$$

$$\begin{aligned}
Z^2 &= \frac{(\bar{Y}_2 - \bar{Y}_1)^2}{s_1^2/n_1 + s_2^2/n_2} \\
&= \frac{(\bar{Y}_2 - \bar{Y}_1)^2}{S_1/[n_1(n_1 - 1)] + S_2/[n_2(n_2 - 1)]}.
\end{aligned}$$

$$W_{\text{ML}} \neq Z^2$$

For any finite n_1, n_2 .

$$W_{\text{Wald}}(s_1^2, s_2^2) = \frac{(\hat{\beta}^*)^2}{s_1^2/n_1 + s_2^2/n_2} = Z^2.$$

The difference vanishes asymptotically as both group sizes grow. Equal group sizes alone do not make the finite-sample statistics equal.

$$\frac{\hat{\sigma}_{g,\text{ML}}^2}{s_g^2} = \frac{n_g - 1}{n_g} \rightarrow 1 \quad n_g \rightarrow \infty.$$

3 Question 2

For $X_j = 0$, let $p_1 = \beta_0$.

For $X_j = 1$, let $p_2 = \beta_0 + \beta^*$.

$$H_0 : p_1 = p_2 \equiv H_0 : \beta^* = 0.$$

Let $G_1 = \{j : X_j = 0\}$ and $G_2 = \{j : X_j = 1\}$. The likelihood is

$$L(\beta_0, \beta^*) = \prod_{j \in G_1} \beta_0^{Y_j} (1 - \beta_0)^{1-Y_j} \prod_{j \in G_2} (\beta_0 + \beta^*)^{Y_j} (1 - \beta_0 - \beta^*)^{1-Y_j}.$$

The log-likelihood is

$$\ell(\beta_0, \beta^*) = \sum_{j \in G_1} [Y_j \log(\beta_0) + (1 - Y_j) \log(1 - \beta_0)] + \sum_{j \in G_2} [Y_j \log(\beta_0 + \beta^*) + (1 - Y_j) \log(1 - \beta_0 - \beta^*)]$$

$$\begin{aligned} \frac{\partial \ell}{\partial \beta_0} &= \sum_{j \in G_1} \frac{Y_j - p_1}{p_1(1 - p_1)} + \sum_{j \in G_2} \frac{Y_j - p_2}{p_2(1 - p_2)}, \\ \frac{\partial \ell}{\partial \beta^*} &= \sum_{j \in G_2} \frac{Y_j - p_2}{p_2(1 - p_2)}. \end{aligned}$$

Setting both scores to zero yields

$$\begin{aligned} \hat{p}_1 &= \frac{1}{n_1} \sum_{j=1}^{n_1+n_2} (1 - X_j) Y_j, \\ \hat{p}_2 &= \frac{1}{n_2} \sum_{j=1}^{n_1+n_2} X_j Y_j, \\ \hat{\beta}_0 &= \hat{p}_1, \\ \hat{\beta}^* &= \hat{p}_2 - \hat{p}_1. \end{aligned}$$

The Fisher information matrix is

$$I(\beta_0, \beta^*) = \begin{pmatrix} \frac{n_1}{p_1(1-p_1)} + \frac{n_2}{p_2(1-p_2)} & \frac{n_2}{p_2(1-p_2)} \\ \frac{n_2}{p_2(1-p_2)} & \frac{n_2}{p_2(1-p_2)} \end{pmatrix}.$$

With the equation of inversion

$$V = \begin{pmatrix} a & b \\ b & c \end{pmatrix}^{-1} = \frac{1}{ac - b^2} \begin{pmatrix} c & -b \\ -b & a \end{pmatrix}, \quad ac - b^2 \neq 0.$$

$$\begin{aligned} I(\beta_0, \beta^*)^{-1} &= \frac{p_1(1-p_1)p_2(1-p_2)}{n_1 n_2} \begin{pmatrix} \frac{n_2}{p_2(1-p_2)} & -\frac{n_2}{p_2(1-p_2)} \\ -\frac{n_2}{p_2(1-p_2)} & \frac{n_1}{p_1(1-p_1)} + \frac{n_2}{p_2(1-p_2)} \end{pmatrix} \\ &= \begin{pmatrix} \frac{p_1(1-p_1)}{n_1} & -\frac{p_1(1-p_1)}{n_1} \\ -\frac{p_1(1-p_1)}{n_1} & \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2} \end{pmatrix}. \end{aligned}$$

Hence, the variance function is:

$$\widehat{\text{Var}}(\hat{\beta}^*) = \frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}.$$

The Wald statistic for $(H_0: \beta^*=0)$ is

$$\begin{aligned} W &= \frac{(\hat{\beta}^*)^2}{\widehat{\text{Var}}(\hat{\beta}^*)} \\ &= \frac{(\hat{p}_2 - \hat{p}_1)^2}{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2} \\ &\equiv Z^2 \end{aligned}$$

This is exactly the $W \equiv Z^2$ statistic specified in the question. Under H_0 and the usual large-sample regularity conditions, W has an asymptotic χ_1^2 distribution, so as Z^2 .

4 Question 3

5 Question 4

6 Question 5

Let $Y_j \sim \text{Binomial}(n_j, p_j)$ independently for groups $j = 1, 2$. Under the log-link model,

$$p_1 = \exp(\beta_0), \quad p_2 = \exp(\beta_0 + \beta^*).$$

We test $H_0 : \beta^* = 0$ against $H_A : \beta^* \neq 0$. This is equivalent to testing $p_1 = p_2$. Define

$$\hat{p}_1 = \frac{Y_1}{n_1}, \quad \hat{p}_2 = \frac{Y_2}{n_2}, \quad \tilde{p} = \frac{Y_1 + Y_2}{n_1 + n_2}.$$

The unrestricted maximum likelihood estimate is $\hat{\beta}^* = \log(\hat{p}_2/\hat{p}_1)$. The Fisher information matrix and its inverse are

$$I(\beta_0, \beta^*) = \begin{pmatrix} \frac{n_1 p_1}{1 - p_1} + \frac{n_2 p_2}{1 - p_2} & \frac{n_2 p_2}{1 - p_2} \\ \frac{n_2 p_2}{1 - p_2} & \frac{n_2 p_2}{1 - p_2} \end{pmatrix}$$

and

$$I(\beta_0, \beta^*)^{-1} = \begin{pmatrix} \frac{1 - p_1}{n_1 p_1} & -\frac{1 - p_1}{n_1 p_1} \\ -\frac{1 - p_1}{n_1 p_1} & \frac{1 - p_1}{n_1 p_1} + \frac{1 - p_2}{n_2 p_2} \end{pmatrix}.$$

6.0.1 a. Wald test

The lower-right entry of I^{-1} gives the estimated variance of $\hat{\beta}^*$. Therefore, the Wald statistic is

$$W = \frac{[\log(\hat{p}_2/\hat{p}_1)]^2}{\frac{1 - \hat{p}_1}{n_1 \hat{p}_1} + \frac{1 - \hat{p}_2}{n_2 \hat{p}_2}}.$$

6.0.2 b. Score test

Under H_0 , both groups have the common probability $\tilde{p}$. After accounting for the nuisance parameter β_0 , the efficient score and its information are

$$U_{\text{eff}} = \frac{Y_2 - n_2 \tilde{p}}{1 - \tilde{p}}, \quad I_{\text{eff}} = \frac{n_1 n_2 \tilde{p}}{(n_1 + n_2)(1 - \tilde{p})}.$$

Hence the score statistic simplifies to

$$S = \frac{U_{\text{eff}}^2}{I_{\text{eff}}} = \frac{(\hat{p}_2 - \hat{p}_1)^2}{\tilde{p}(1 - \tilde{p})(1/n_1 + 1/n_2)}.$$

6.0.3 c. Likelihood ratio test

The unrestricted model uses $\hat{p}_1$ and $\hat{p}_2$; the model under H_0 uses $\tilde{p}$ for both groups. Twice the difference between their maximized log likelihoods is

$$G^2 = 2 \sum_{j=1}^2 \left[Y_j \log \left(\frac{\hat{p}_j}{\tilde{p}} \right) + (n_j - Y_j) \log \left(\frac{1 - \hat{p}_j}{1 - \tilde{p}} \right) \right]$$

Under H_0 , each of W , S , and G^2 has an asymptotic χ_1^2 distribution. At significance level 0.05, reject H_0 when the chosen statistic exceeds 3.841. The formulas above assume interior estimated probabilities ($0 < \hat{p}_1, \hat{p}_2, \tilde{p} < 1$).